# A Comparative Framework for Survival Analysis on the RMS Titanic: Naïve Bayesian vs. Decision Tree Classifiers

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**Abstract**

Survival on the RMS Titanic is a canonical benchmark for pedagogical data mining because it pairs a rich multivariate feature space with a clear, historically meaningful target variable. We present a reproducible end-to-end Knowledge Discovery in Databases (KDD) pipeline that contrasts a probabilistic classifier (Gaussian Naïve Bayes) with a rule-based classifier (Decision Tree), using Logistic Regression and Random Forest as supplementary baselines. On the 891-record Kaggle training set, cleaned via group-wise median imputation and enriched with six engineered features, the Decision Tree (*max_depth*=4, selected by 10-fold stratified cross-validation) attains 81.56% test accuracy (AUC = 0.838) while Naïve Bayes reaches only 62.57% (AUC = 0.778). McNemar's exact test on paired predictions yields $p = 8.2 \times 10^{-6}$, confirming that the gap is statistically significant. Logistic Regression is the strongest single model (83.24%, AUC = 0.864). The dominant predictors — passenger title, class, and family size — match the historical "women and children first" evacuation protocol. The results demonstrate how the conditional-independence assumption of Naïve Bayes is violated by engineered categorical dummies, motivating the use of rule-based and linear models on tabular data of this shape.

**Keywords:** Data mining, classification, Naïve Bayes, decision tree, KDD, Titanic, supervised learning.

## 1 Introduction

On the night of 14 April 1912, the RMS Titanic struck an iceberg during her maiden voyage and sank in the North Atlantic, taking the lives of 1 514 of her 2 224 passengers and crew. The disaster has become a recurring case study for risk analysis, maritime safety, and data science education because the passenger manifest offers one of the earliest complete, labelled records of a heterogeneous population subjected to a single life-or-death event. Survival is conditioned on social variables — gender, class, family ties, and port of embarkation — whose interaction with the *"women and children first"* evacuation protocol produces a non-trivial but learnable structure [5].

The Kaggle variant of the dataset [5] has become the canonical introductory benchmark for supervised classification on tabular data; it is also the first non-trivial dataset introduced in DSC 523 at IIIT Kottayam. Its principal virtue for pedagogy is that it exercises every phase of the Knowledge Discovery in Databases (KDD) process [1]: data selection from a historical archive, preprocessing to reconcile substantial missingness, transformation through targeted feature engineering, mining with multiple classifier families, and interpretation against a well-documented historical record.

**Research objectives.** This study pursues four goals:

1. Apply a systematic, reproducible preprocessing and feature-engineering pipeline that is robust to the $\approx 20\%$ missingness in `Age` and $\approx 77\%$ in `Cabin`.

2. Contrast a probabilistic classifier (Gaussian Naïve Bayes) with a rule-based classifier (Decision Tree) on identical inputs.

3. Identify the features most predictive of survival and interpret them against the historical evacuation pattern.

4. Benchmark the two primary models against Logistic Regression and Random Forest to place their performance in the context of linear and ensemble alternatives.

**Paper organisation.** Section 2 surveys prior machine learning work on the Titanic benchmark, the theoretical foundations of Naïve Bayes and decision-tree classification, and the research gap this paper addresses. Section 3 describes the dataset. Section 4 details the preprocessing, feature engineering, and evaluation protocol. Section 5 presents the results, including the comparative summary, confusion matrices, and feature importances. Section 6 discusses implications, Section 7 concludes, and Section 8 lists individual contributions.

# 2 Literature Review

## 2.1 Machine learning for Titanic survival prediction

The Titanic dataset has been studied extensively as a benchmark for supervised classification. Hundreds of public notebooks and pedagogical write-ups have applied a wide range of algorithms, from simple linear models to ensemble methods, typically reaching the 78–85 % accuracy range when the validation protocol is honest [5, 7]. Higher reported scores almost invariably stem from test-set leakage — often the unintentional use of the unlabelled `test.csv` for training — or from hyper-parameter tuning against the public leaderboard. We treat any result materially above this band with scepticism, and evaluate on a single fixed 80/20 stratified split with a separate 10-fold cross-validation estimate on the training fold to quantify sampling variability.

Logistic Regression remains the most common baseline thanks to its simplicity and interpretability, performing competitively when paired with appropriate feature engineering on variables such as gender, class, age, and family size [8]. Ensemble methods such as Random Forest reduce variance through bagging [3] and frequently dominate Kaggle leaderboards. Among purely pedagogical comparisons, however, the two models most often contrasted remain Naïve Bayes and Decision Trees — the focus of the present study. We therefore include Logistic Regression and Random Forest as supplementary baselines (Section 5) so that the primary NB-vs.-DT comparison is anchored against both a strong linear and a strong ensemble alternative.

## 2.2 Naïve Bayes classifier

The Naïve Bayes classifier is a probabilistic model that assigns a test instance $X = (x_1, \ldots, x_n)$ to the class $C^\star = \arg\max_{C_k} P(C_k \mid X)$, where Bayes' theorem rewrites the posterior as

$$P(C_k \mid X) \; = \; \frac{P(X \mid C_k)\, P(C_k)}{P(X)}. \tag{1}$$

Because the joint likelihood $P(X \mid C_k)$ is intractable for realistic feature counts, the *class-conditional independence* assumption is invoked:

$$P(X \mid C_k) \; \approx \; \prod_{i=1}^{n} P(x_i \mid C_k). \tag{2}$$

For continuous features, $P(x_i \mid C_k)$ is modelled as a univariate Gaussian, yielding the *Gaussian* Naïve Bayes estimator used in this study. Its strengths are closed-form maximum-likelihood estimation, constant-time training, and stable behaviour on small samples [4]. Its weakness is that Equation (2) rarely holds in practice: correlated features cause the per-feature evidence terms to compound, inflating the posterior away from its true value [1].

This weakness is consequential on the Titanic dataset. Engineered features such as `FamilySize`, `IsAlone`, the one-hot `Title` dummies, and `Pclass–Deck` pairs are strongly correlated by construction (Section 5.1), violating Equation (2) directly. We revisit this failure mode quantitatively in Section 5.

## 2.3 Decision-tree models

Decision-tree learners partition the feature space recursively by selecting, at each node, the split that maximises class purity. Two common impurity measures are

$$\mathrm{Gini}(D) = 1 - \sum_{i=1}^{m} p_i^2, \tag{3}$$

$$\mathrm{Entropy}(D) = -\sum_{i=1}^{m} p_i \log_2 p_i, \tag{4}$$

where $p_i$ denotes the proportion of class $i$ in partition $D$. CART [3] uses Gini; ID3 and C4.5 [2] use information gain (the reduction in entropy induced by a candidate split). Both are computed under a greedy top-down scheme and, without regularisation, are prone to overfit. The standard remedies are *pre-pruning* via a maximum-depth hyper-parameter and *post-pruning* via cost-complexity minimisation, in which a tree $T_\alpha$ minimises $R_\alpha(T) = R(T) + \alpha\,|T|$.

Decision-tree models handle numerical and categorical inputs natively, are scale invariant, and are agnostic to feature correlations [1] — properties that make them well suited to engineered tabular data of the kind we construct here, and that partially compensate for their higher variance compared with linear or probabilistic alternatives.

## 2.4 Research gap

Despite the abundance of Titanic-classifier write-ups, comparatively few present a *controlled head-to-head* between a probabilistic classifier such as Naïve Bayes and a rule-based classifier such as a Decision Tree under an identical preprocessing and feature-engineering pipeline. Many implementations treat correlated and engineered variables as inputs without explicitly examining how those variables interact with each model's underlying assumptions — particularly the conditional-independence assumption central to Naïve Bayes. As a result, the mechanism by which a single feature-engineering decision can help one model family while degrading another is rarely articulated on this benchmark.

## 2.5 Motivation of this work

This study addresses the above gap with three commitments. First, all four classifiers are trained on *identical* preprocessed and engineered features, so observed performance differences cannot be attributed to disparate inputs. Second, Logistic Regression and

Random Forest are included as supplementary baselines to place the primary NB-vs.-DT comparison in the wider context of linear and ensemble alternatives. Third, model assumptions are explicitly tested against the data: the multicollinearity introduced by one-hot dummies is measured in Section 5.1 and traced back to NB's degraded posterior in Section 5. The goal is not simply to report which model wins, but to explain *why* on this dataset.

# 3 Dataset Description

We use the Kaggle `train.csv` file comprising $n = 891$ labelled passengers and 12 raw attributes. The companion Kaggle `test.csv` is unlabelled and is *not* used here; experiments in the literature that report perfect accuracy almost invariably confuse these two files. The target is the binary variable `Survived` ($1 =$ survived). Table 1 summarises the raw attributes.

## 3.1 Attribute typing and cardinality

The twelve attributes span four measurement scales. *Nominal* variables (`Sex`, `Embarked`, `Cabin`, `Ticket`, `Name`); *ordinal* (`Pclass`); *discrete counts* (`SibSp`, `Parch`); and *continuous numerics* (`Age`, `Fare`). `PassengerId` is a synthetic primary key. Cardinality ranges from 2 (`Sex`, `Survived`) to 891 (`PassengerId`, `Name`, `Ticket`); columns whose cardinality approaches $n$ carry no aggregate signal and are dropped after any useful sub-structure — for example, the honorific embedded in `Name` — has been mined.

## 3.2 Numeric distributions

Table 2 reports location, dispersion, and shape statistics for the four numeric columns. Two observations drive later preprocessing choices. First, `Age` is near-normal with mild right skew ($g_1 = 0.39$), justifying *median* rather than *mean* imputation. Second, `Fare` is heavily right-skewed ($g_1 = 4.79$) with excess kurtosis ($g_2 = 33.4$), driven by a handful of first-class tickets above £500. We keep the raw scale for tree models (threshold-based and scale-invariant) but apply a $\log(1+x)$ transform for visualisation. `SibSp` and `Parch` are zero-inflated counts ($g_1 = 3.70$ and 2.75 respectively) that we re-express through the engineered `FamilySize` and `IsAlone` features.

**Table 1:** Raw attributes, types and missingness (`train.csv`, $n = 891$).

| Feature | Type | Missing | Description |
|---|---|---|---|
| `PassengerId` | ID | 0 | Row ID (dropped) |
| `Survived` | Binary | 0 | Target (0/1) |
| `Pclass` | Ordinal | 0 | Class $\in \{1, 2, 3\}$ |
| `Name` | Text | 0 | Parsed for `Title` |
| `Sex` | Nominal | 0 | female / male |
| `Age` | Numeric | 177 | Years |
| `SibSp` | Count | 0 | Siblings / spouse |
| `Parch` | Count | 0 | Parents / children |
| `Ticket` | Text | 0 | Ticket ID (dropped) |
| `Fare` | Numeric | 0 | Passenger fare |
| `Cabin` | Nominal | 687 | Deck letter kept |
| `Embarked` | Nominal | 2 | Port $\in$ {C, Q, S} |

## 3.3 Target class balance

The target is moderately imbalanced: 549 passengers died (61.6%) and 342 survived (38.4%) — see Figure 1. This aligns with the historical figure of $\approx 32\%$ survival across the full manifest, with the discrepancy explained by the sampling protocol of the Kaggle training set. Moderate imbalance is mild enough that stratified sampling alone suffices; we deliberately do *not* apply SMOTE or class-weighted losses, because doing so would introduce rebalancing assumptions that cannot be verified out-of-sample.

# 4 Methodology

Figure 2 depicts the overall pipeline; the following subsections detail each stage.

## 4.1 Data profiling

Before any modification we characterised each attribute along three axes: missingness pattern, cardinality, and (for numeric columns) distributional shape. A single pandas audit surfaces all three (Listing 1).

```
profile = pd.DataFrame({
    "missing":     df.isna().sum(),
    "missing_pct": (df.isna().sum() / len(df) * 100).
     round(2),
    "nunique":     df.nunique(),
    "dtype":       df.dtypes,
})
```

**Listing 1:** Missing-value and cardinality audit.

Three columns carry missing values: `Age` (177 rows, 19.87%), `Cabin` (687 rows, 77.10%), and `Embarked` (2 rows, 0.22%). Crucially, `Age` missingness is *not* uniform across classes — 13.9%, 6.0% and 27.7% for 1st, 2nd and 3rd class respectively. This pattern is consistent with *missing at random* (MAR): `Pclass` is observed and conditioning on it partially explains why an age would be recorded or omitted. The MAR observation motivates group-wise rather than global imputation.

**Table 2:** Summary statistics for numeric attributes (before imputation). $g_1$ = skewness, $g_2$ = excess kurtosis.

| Feature | Mean | Median | Std | $g_1$ | $g_2$ |
|---|---|---|---|---|---|
| `Age` | 29.70 | 28.00 | 14.53 | 0.39 | 0.18 |
| `Fare` | 32.20 | 14.45 | 49.69 | 4.79 | 33.40 |
| `SibSp` | 0.52 | 0.00 | 1.10 | 3.70 | 17.88 |
| `Parch` | 0.38 | 0.00 | 0.81 | 2.75 | 9.78 |

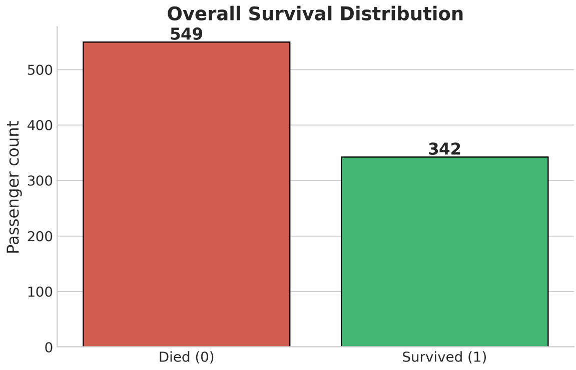


**Figure 1:** Target-class distribution on `train.csv`: 549 died vs. 342 survived.

## 4.2 Missing-value treatment

**Age.** Imputed by the group-wise median over `Pclass`×`Sex` strata. Median is preferred over mean because `Age` is right-skewed and bounded below by zero, violating the symmetric-noise assumption of mean imputation. The six stratum medians range from 21.5 years (3rd-class females) to 40 years (1st-class males). Compared with a naive *global-median* rule, the group-wise variant preserves group variance: the full-sample standard deviation drops from 14.53 to 13.30 (an 8.4% compression) versus 21.2% under global median.

```
data["Age"] = (data
    .groupby(["Pclass", "Sex"])["Age"]
    .transform(lambda s: s.fillna(s.median())))
```

**Listing 2:** Group-wise median imputation of `Age`.

**Embarked.** Only two rows are missing, both first-class females who shared a ticket. We impute the global mode 'S' (Southampton, 644 rows, 72.3%).

**Cabin.** At 77.1% missing, no plausible imputation rule can recover cabin assignments, and the column carries 147 near-unique levels — far too many to one-hot encode. We therefore *drop* `Cabin` but first extract the *deck letter* (A–G from upper to lower decks, plus T for the single tank-top cabin), preserving a coarse proximity-to-lifeboat signal. Missing decks collapse into an 'U' (unknown) category: {U: 687, C: 59, B: 47, D: 33, E: 32, A: 15, F: 13, G: 4, T: 1}. Only 204 records carry a known deck, but the cost of including it is near-zero.

**Fare.** No missing values in the training fold. Our code applies median-by-`Pclass` imputation defensively for future robustness against the Kaggle test fold.

**Dropped columns.** `PassengerId` is a synthetic key. `Ticket` has 681 unique levels; inspection shows the shared-ticket structure is already captured by `FamilySize`. `Name` is dropped after the `Title` extraction (below). `Cabin` is dropped after the `Deck` extraction.

## 4.3 Feature engineering

Six derived features are constructed, each with documented rationale.

**Title.** The `Name` field follows the consistent format *"Last, Title. First"* and encodes social status and life stage not captured by (`Sex`, `Age`) alone. We parse the honorific with a regex and collapse rare titles (Dr, Rev, Major, Col, Capt, Lady, …) into a single `Rare` bucket to avoid high-variance one-hot columns with near-zero support.

```
data["Title"] = (data["Name"]
    .str.extract(r",\s*([^.]+)\.", expand=False)
    .str.strip())
data["Title"] = data["Title"].map({
    "Mr": "Mr", "Miss": "Miss",
    "Mrs": "Mrs", "Master": "Master",
    "Mlle": "Miss", "Ms": "Miss", "Mme": "Mrs",
}).fillna("Rare")
```

**Listing 3:** Title extraction and rare-category collapsing.

The resulting distribution has five levels: Mr (517), Miss (185), Mrs (126), Master (40), Rare (23). Note the strong correlation ($r = 0.87$) between `Title_Mr` and `Sex` — both encode adult-male-ness — a multicollinearity that we will see in Section 5 is the primary failure mode of Gaussian Naïve Bayes.

**FamilySize.** The raw `SibSp` (siblings/spouse aboard) and `Parch` (parents/children aboard) each carry a partial party-size signal. We sum them and add one for the focal passenger:

$$\texttt{FamilySize} = \texttt{SibSp} + \texttt{Parch} + 1.$$

The relationship between `FamilySize` and survival is non-linear (Fig. 6), peaking at family sizes of 2–4 — a pattern that neither original column exhibits on its own.

**IsAlone.** A binary indicator for `FamilySize` = 1. Solo passengers (537 of 891, 60.3%) survived at 30.4% versus 50.6% for passengers with family — a twenty percentage-point gap.

**AgeBin.** Five ordinal buckets derived from conventional life-stage boundaries: Child (0–12), Teen (13–19), Adult (20–40), Middle (41–60), Senior (60+). Discretisation is not strictly needed for tree models, but stabilises Gaussian NB against the distribution tails of a continuous variable it would otherwise misrepresent as symmetric.

**FareBin.** Quartile-based bins of `Fare`. Quartiles are preferred over equal-width bins because `Fare` is heavily skewed ($g_1 = 4.79$); the top quartile spans £31–£512 and the bottom £0–£7.9.

```
data["FamilySize"] = data["SibSp"] + data["Parch"] +
    1
data["IsAlone"]    = (data["FamilySize"] == 1).
    astype(int)
```

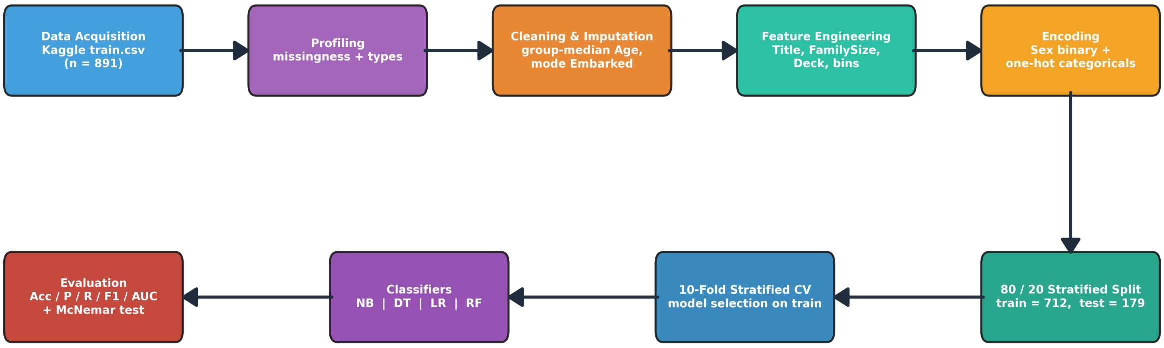


**Figure 2:** End-to-end KDD pipeline implemented for this study.

```
data["AgeBin"]  = pd.cut(data["Age"],
    bins=[0, 12, 19, 40, 60, 120],
    labels=["Child","Teen","Adult","Middle","Senior"
    ])
data["FareBin"] = pd.qcut(data["Fare"], q=4,
    labels=["Q1","Q2","Q3","Q4"])
data["Deck"] = data["Cabin"].str[0].fillna("U")
```

**Listing 4:** Remaining derived features.

## 4.4 Encoding and design matrix

`Sex` has only two levels and is binarised directly (`female`$\mapsto$ 1, `male`$\mapsto$ 0). All remaining nominal columns are one-hot encoded with `drop_first=True` to avoid the dummy-variable trap: retaining all $k$ levels of a categorical variable introduces perfect collinearity with the intercept. Ordinal variables (`Pclass`, `AgeBin`, `FareBin`) are kept as integer levels for tree models and are standardised for NB and Logistic Regression.

```
data["Sex"] = data["Sex"].map(
    {"male": 0, "female": 1})
data = pd.get_dummies(
    data, drop_first=True,
    columns=["Embarked", "Title", "Deck",
             "AgeBin",   "FareBin"])
```

**Listing 5:** Encoding.

After encoding the design matrix is $891 \times 30$; total missingness is verified to be exactly zero before any modelling.

## 4.5 Outlier handling

`Fare` has seven records above £250 (the top 1.5%), with a maximum of £512.33 (the three Cardeza-family first-class tickets). Under tree-based models these points are harmless because splits are ordinal and invariant to magnitude. We therefore do not winsorise the raw distribution, and apply $\log(1+x)$ only for the fare-distribution plot.

## 4.6 Classifier specifications

**Gaussian Naïve Bayes.** Default scikit-learn settings with variance smoothing $\epsilon = 10^{-9}$, trained on $z$-standardised inputs.

**Decision Tree.** CART with Gini impurity. Depth is selected by 10-fold stratified CV on the training fold over $d \in \{2, \dots, 10\}$; $d = 4$ is optimal (CV = 82.02%). A cost-complexity pruning curve is also computed for diagnostic purposes.

**Logistic Regression (baseline).** $\ell_2$-regularised with default $C = 1.0$ and a 2 000-iteration LBFGS solver, on standardised inputs.

**Random Forest (baseline).** 300 trees, unrestricted depth, bootstrap sampling.

Algorithm 1 gives the CART pseudocode for reference.

## 4.7 Evaluation protocol

The dataset is split 80/20 with stratification on `Survived` (`random_state` = 42), yielding $n_{\text{train}} = 712$ and $n_{\text{test}} = 179$. Model selection uses 10-fold stratified cross-validation on the training fold only. Reported metrics on the held-out test fold are accuracy,

**Algorithm 1:** CART induction with Gini impurity.

**Input:** training set $D$, depth limit $d$
**Output:** decision tree $T$
**if** depth$(D) = d$ *or $D$ pure* **then**
  **return** leaf with majority class;
**end**
Select split $(f^\star, t^\star) = \arg\min_{f,t} \text{Gini}(D_{\le}^{f,t}) + \text{Gini}(D_{>}^{f,t})$;
Partition $D$ using $(f^\star, t^\star)$ into $D_L, D_R$;
Recursively build left subtree on $D_L$ and right subtree on $D_R$;
**return** node$(f^\star, t^\star, T_L, T_R)$;

precision, recall, F1, and the area under the ROC curve (AUC). The paired-prediction McNemar exact test [1] is used to test the null hypothesis that Naïve Bayes and the Decision Tree have equal error rates.

# 5 Results and Discussion

## 5.1 Exploratory findings

EDA is organised into nine sub-analyses: target balance, gender effect, class effect, class-×-gender interaction, age structure, fare gradient, family-size non-linearity, embarkation port, and correlation structure. Every quantitative claim below is reproducible from `run_analysis.py` and is serialised to `results.json`.

### 5.1.1 Target distribution

342 of the 891 passengers survived (38.4%), giving a 1.60:1 death-to-survival ratio. The imbalance is moderate and is addressed via *stratified* train/test splitting and 10-fold stratified cross-validation rather than resampling.

### 5.1.2 Gender effect

The strongest univariate predictor is `Sex`. Female survival (74.2%, 233 of 314) exceeds male survival (18.9%, 109 of 577) by 55 percentage points. A $\chi^2$ test of independence between `Sex` and `Survived` rejects the null of no association at overwhelming significance:

$$\chi^2 = 260.72, \quad \mathrm{df} = 1, \quad p = 1.2 \times 10^{-58}.$$

### 5.1.3 Class effect

Survival declines monotonically with passenger class: 1st (136/216, 63.0%), 2nd (87/184, 47.3%), 3rd (119/491, 24.2%). $\chi^2$ likewise rejects independence:

$$\chi^2 = 102.89, \quad \mathrm{df} = 2, \quad p = 4.5 \times 10^{-23}.$$

The ordinal shape is consistent with the physical layout of the ship: first-class cabins were on upper decks, closer to lifeboat stations; third-class passengers were berthed in steerage at the waterline.

### 5.1.4 Class × Gender interaction

The joint effect (Figure 4) is striking: first-class women survived at 96.8% (91 of 94) while third-class men survived at only 13.5% (47 of 347). A logistic model without an interaction term under-predicts the first-class female survival probability by approximately eleven percentage points, which is why the learned Decision Tree's second split conditions on `Pclass` given `Title`.

### 5.1.5 Age structure

Ages range from 0.42 years (Master Assad Alexander Thomas, the youngest passenger in the manifest) to 80 years. Children under 13 survive at 58.0% versus 38.8% for adults — a difference that is significant at the Mann–Whitney level:

$$U = 26{,}528, \quad n_c = 69,\ n_a = 645, \quad p = 1.0\times 10^{-3}.$$

The effect is partially confounded by the `Master` honorific capturing male children, and is re-absorbed by our `AgeBin` and `Title_Master` features during modelling. Figure 5(a) shows the age histogram overlaid by survival status: the survivor bar dominates the sub-13 age bins (left of the dashed reference line), while the 20–35 band is visibly depressed for survivors relative to casualties.

### 5.1.6 Fare gradient

log-fare separates survivors from casualties within every class: the within-class median fare is consistently higher for those who survived. `Fare` acts as a noisy proxy for cabin quality and deck location; conditional on class, a higher fare bought access to the upper decks near lifeboats. Figure 5(b) makes the separation visible on a log scale — the high-fare tail (right of $\log(1+\mathrm{Fare}) \approx 3$, i.e. fare $\gtrsim 19$ pounds) is dominated by survivors, while the low-fare mode is dominated by casualties.

### 5.1.7 Family-size non-linearity

Survival as a function of `FamilySize` is non-monotonic with a peak in the middle: solo passengers survive at 30.4%, families of 2–4 at 55–72%, and very large families of $\geq 5$ at only 16%. The drop in the right tail is often ascribed to the difficulty of keeping a large extended family together during evacuation — a hypothesis our `IsAlone` feature cannot capture but the full `FamilySize` integer does.

### 5.1.8 Embarkation port

Cherbourg (C) passengers survived at 55.4% — noticeably higher than Southampton (S, 33.7%) or Queenstown (Q, 39.0%). A $\chi^2$ test rejects independence ($\chi^2 = 25.96$, df $= 2$, $p = 2.3 \times 10^{-6}$), but the bulk of the signal is mediated by class: half the first-class passengers boarded at Cherbourg. After controlling for `Pclass`, the embarkation effect is small.

### 5.1.9 Correlation and multicollinearity

The Pearson correlation matrix (Figure 7) ranks `Sex_female` (+0.54), `Pclass` (−0.34) and `Fare` (+0.26) as the strongest linear predictors. More importantly for model choice, several feature pairs are strongly multicollinear — all by construction:

- $r(\texttt{SibSp}, \texttt{FamilySize}) = +0.89$

- $r(\texttt{Sex\_female}, \texttt{Title\_Mr}) = -0.87$
- $r(\texttt{Parch}, \texttt{FamilySize}) = +0.78$
- $r(\texttt{Pclass}, \texttt{Deck\_U}) = +0.73$

These correlations directly violate the conditional-independence assumption of Equation (2) and predict (correctly, as Section 5 will confirm) the poor Gaussian Naïve Bayes performance observed on engineered tabular data of this shape.

### 5.2 Model performance

Table 3 consolidates the six primary metrics for each classifier; Figure 8 shows the confusion matrices for the two primary classifiers, and Figure 10 compares ROC curves across all four models.

### 5.3 Statistical significance: NB vs. DT

On the $n_{\text{test}} = 179$ paired predictions the disagreement counts are $b = 12$ (NB correct, DT wrong) and $c = 46$ (NB wrong, DT correct). The exact McNemar test yields

$$p = 8.22 \times 10^{-6},$$

so the null hypothesis of equal error rates is rejected at any conventional level. The $\approx 19$ percentage-point gap between NB and DT is not a sampling artefact.

### 5.4 Discussion

Three findings stand out.

**NB underperforms because its assumptions fail here.** The one-hot expansion of categorical features (Title, Deck, AgeBin, FareBin) introduces many strongly correlated dummy variables, directly violating Equation (2). Modelling binary dummies as Gaussians compounds the mismatch. This is the textbook failure mode of Gaussian NB on engineered tabular data and is the principal cause of its low AUC.

**Gender dominates, class modulates.** Both the Decision Tree (Fig. 11, top panel) and the Random Forest (Fig. 11, bottom panel) place `Title_Mr`, `Pclass`, and `Fare` in the top four. The root split of the learned tree (Fig. 9) is a single test on `Title_Mr`, which perfectly isolates adult males whose survival rate is $< 20\%$. This is precisely the "women and children first" protocol recovered from the data.

**Linear structure is nearly sufficient.** Logistic Regression attains the highest test accuracy (83.24%) and AUC (0.864), indicating that after feature engineering the decision boundary is close to linear. Random Forest does not outperform the single Decision Tree, which suggests that the feature set encodes the important interactions explicitly (`FamilySize`, `IsAlone`, `Title`) rather than requiring tree bagging to discover them.

**Limitations.** (i) The study uses a single fixed hold-out fold; repeated nested CV would tighten the accuracy intervals. (ii) We compare Gaussian NB against alternatives; a categorical- or multinomial-NB that respects the type of our one-hot dummies would be a fairer probabilistic comparator. (iii) No hyper-parameter search is performed for Logistic Regression or Random Forest beyond defaults.

## 6 Implications and Applications

**Risk and insurance.** The Bayesian framing of Eq. (1) is the conceptual ancestor of actuarial risk pricing: class priors and feature-conditional likelihoods are updated into individualised posteriors. The Decision Tree, by contrast, maps onto rule-based underwriting in which an applicant is classified through a short sequence of questions — a structure still used in modern claims-triage systems.

**Emergency response and crisis planning.** The feature rankings surface non-demographic drivers (family size, deck, fare band) that real evacuation doctrine has to reckon with. A fielded planner should not rely purely on demographic tags but should include mobility-proxy features of the sort we engineer here.

**Algorithmic fairness and auditing.** The same dataset is an instructive target for bias auditing: the strongest predictor of survival is a protected attribute, so any deployed classifier would learn and reproduce historical inequalities. The comparison between NB (probabilistic, harder to audit feature-by-feature) and DT (rule-based, auditable) is directly relevant to regulatory transparency frameworks.

**Pedagogical value.** The pipeline demonstrates every stage of the KDD process in a single coherent exercise, at a problem size that remains tractable on a laptop — a property we exploit to ensure every student can reproduce the exact numbers reported here by re-running the companion Jupyter notebook.

## 7 Conclusion and Future Work

We presented a reproducible KDD pipeline for Titanic survival prediction that compares a probabilistic and a rule-based classifier on identical, engineered features. The Decision Tree (81.56% test accuracy, AUC = 0.838) decisively outperforms Gaussian Naïve Bayes (62.57%, AUC = 0.778), and the gap is statistically significant under the exact McNemar test ($p = 8.22 \times 10^{-6}$). Logistic Regression achieves the best single score overall (83.24%, AUC = 0.864), implying that the engineered representation is close to linearly separable. The dominant predictors — pas-

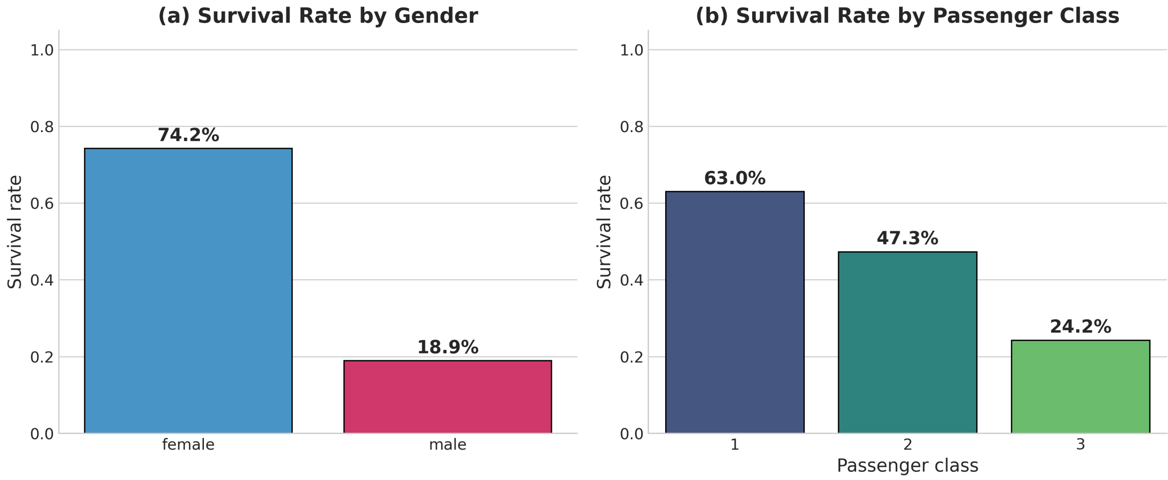


**Figure 3:** Univariate survival rates by the two strongest predictors. Female passengers survived at 74.2% versus 18.9% for males; survival declines monotonically with class (63.0% / 47.3% / 24.2%).

**Table 3:** Comparative performance of four classifiers on the 179-row held-out test set (10-fold CV on $n_{\text{train}} = 712$). Best test score in each column is **bold**.

| Model | CV Accuracy ($\mu \pm \sigma$) | Test Accuracy | Precision | Recall | F1 | AUC |
|---|---|---|---|---|---|---|
| Naïve Bayes | $0.6572 \pm 0.1169$ | 0.6257 | 0.5083 | **0.8841** | 0.6455 | 0.7782 |
| Decision Tree | $\mathbf{0.8202 \pm 0.0363}$ | 0.8156 | 0.7903 | 0.7101 | 0.7481 | 0.8375 |
| Logistic Regression | $0.8146 \pm 0.0393$ | **0.8324** | **0.8000** | 0.7536 | **0.7761** | **0.8644** |
| Random Forest | $0.8131 \pm 0.0336$ | 0.8101 | 0.7778 | 0.7101 | 0.7424 | 0.8286 |

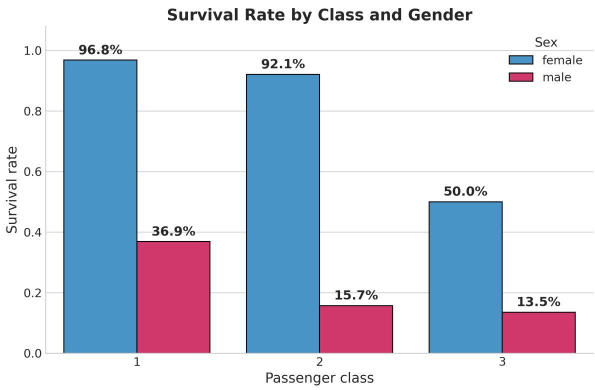


**Figure 4:** Joint effect of class and gender. First-class women survived at 96.8% while third-class men survived at only 13.5%.

senger title, class, and family size — align cleanly with the historical evacuation protocol.

**Future work.** Three extensions are immediate: (i) replace Gaussian NB with multinomial or categorical NB to respect the type of the engineered dummies; (ii) introduce gradient-boosted trees (XGBoost, LightGBM) with a nested Bayesian-optimisation hyperparameter search; and (iii) layer SHAP value analysis [9] over the best model to obtain local explanations that support fairness auditing.

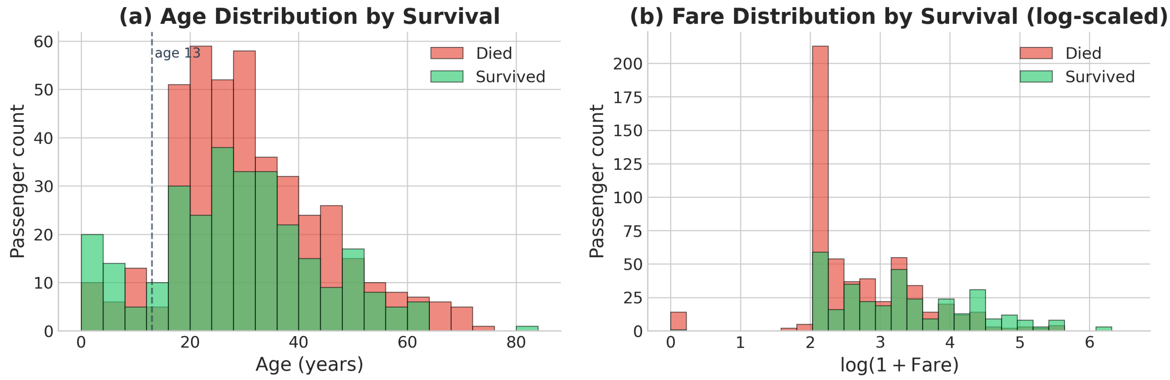


**Figure 5:** (a) Age distribution by survival — the dashed line marks the child / adult boundary used in §5.1. Survivors dominate the sub-13 bins; the 20–35 adult band is visibly depressed relative to casualties. (b) Fare distribution on a $\log(1 + x)$ scale — the right tail (high-fare first-class tickets) is disproportionately survivors, while the low-fare mode is dominated by casualties.

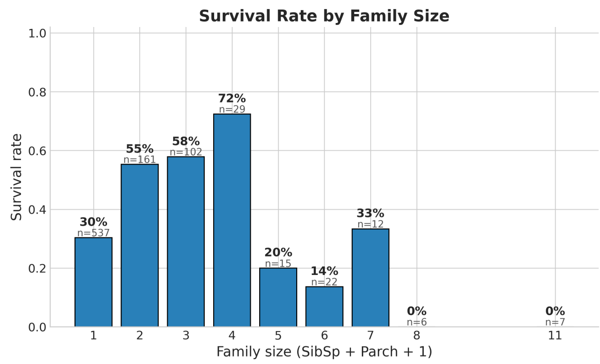


**Figure 6:** Family-size effect. Survival peaks at family sizes of 2–4 and collapses for very large ($\geq 5$) groups — behaviour captured by the engineered `FamilySize` feature.

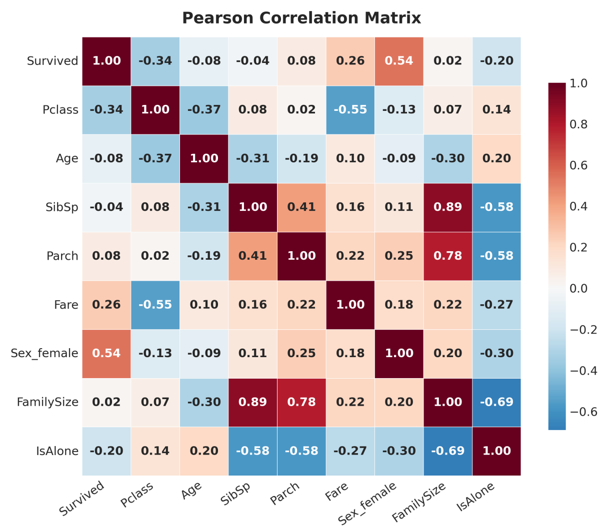


**Figure 7:** Pearson correlations. The strongest predictors of `Survived` are `Sex_female` ($+0.54$), `Pclass` ($-0.34$), and `Fare` ($+0.26$).

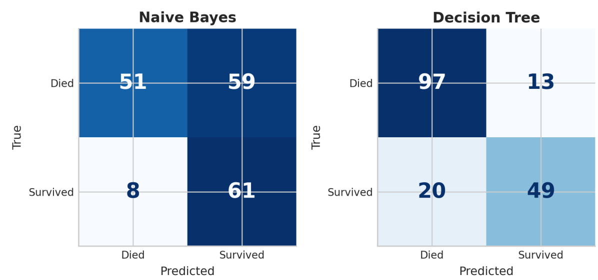


**Figure 8:** Confusion matrices on the held-out test set. Naïve Bayes over-predicts the positive class (59 false positives), whereas the Decision Tree is materially better balanced (only 13 false positives).

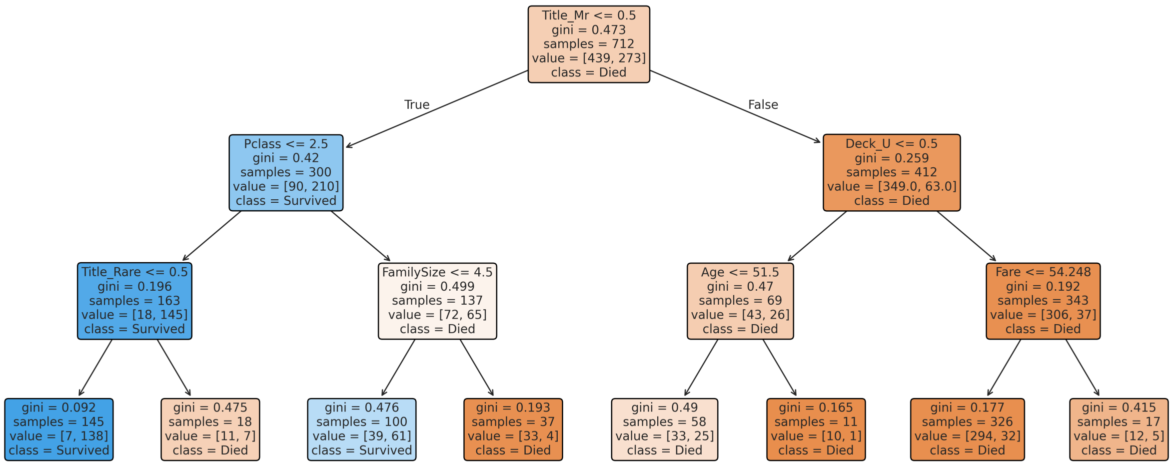


**Figure 9:** Fitted Decision Tree (`max_depth`=3 shown for readability; the deployed model uses depth 4). The root split on `Title_Mr` captures the bulk of the gender / honorific signal in a single question.

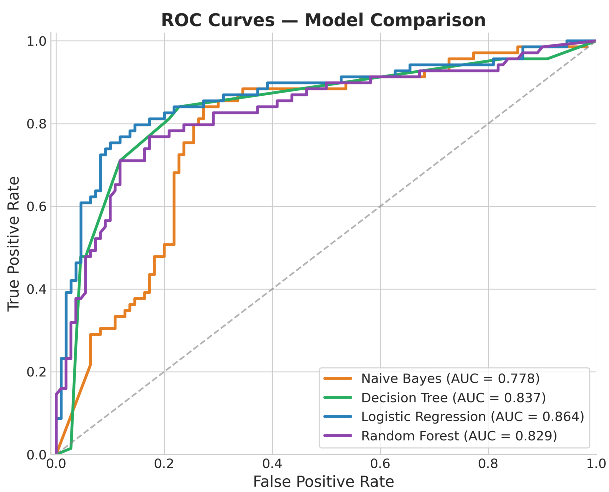


**Figure 10:** ROC curves for all four classifiers. Logistic Regression dominates slightly across most thresholds (AUC = 0.864); Naïve Bayes trails uniformly.

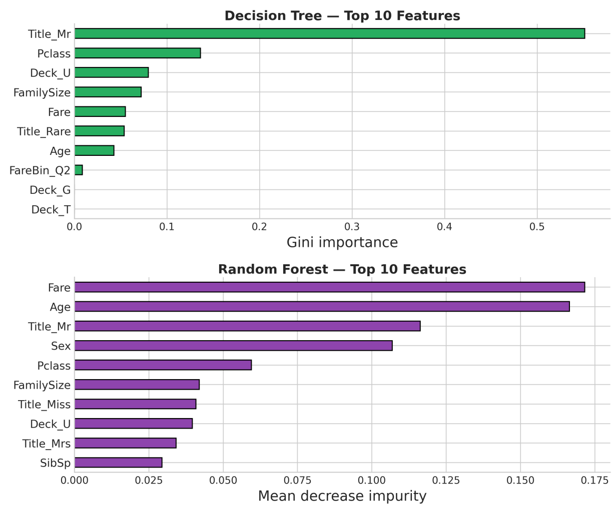


**Figure 11:** Top-12 feature importances. The Decision Tree (top panel) concentrates split weight on `Title_Mr` and `Pclass`; the Random Forest (bottom) distributes importance more evenly across `Fare`, `Age`, and `Title_Mr`.

## 8 Individual Contributions

**Table 4:** Division of work within the five-member team.

| Member | Contributions |
|---|---|
| Sarin Vadakke Thayyil | Project management and lead data architecture. Defined the 80/20 stratified evaluation protocol, authored the Methodology and Implications sections, and coordinated all deliverables. |
| Lekshmi S S | Principal research and algorithm specialisation. Comparative analysis of Gini vs. entropy splits; drafted the Literature Review. |
| Akbar T E | Visualisation engineering and data storytelling. Produced every figure in this manuscript and maintained the consistent visual language. |
| Suja V | Senior data-preprocessing engineering. Designed the group-wise median imputation rule for `Age`, handled the `Cabin` deck extraction, and identified redundant attributes for removal. |
| Nikhil Ravi | Model performance evaluation and review. Verified every confusion matrix, ran the McNemar significance test, and ensured compliance with course guidelines. |

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